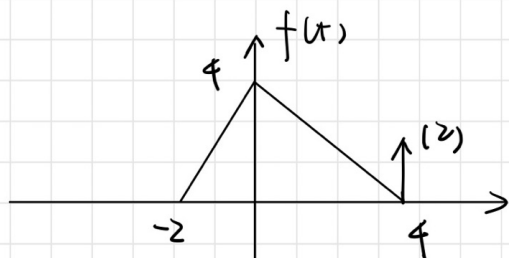


1. 求  $f(-2t-2)$



2.  $\frac{dr(t)}{dt} + r(t) + \int_{-\infty}^t r(\tau) d\tau = e(t)$

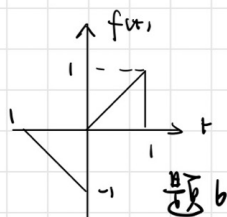
时变? 线性?

3.  $H(p) = \frac{p+3}{p^2+6p+9}$   $r(0^-)=1$ ,  $r'(0^-)=1$ , 求  $r_{zi}$  和  $h(t)$

4.  $f_1(t) = \varepsilon(t+2) - \varepsilon(t+3)$  求  $f_1(t) * f_2(t)$  时域波形  
 $f_2(t) = \varepsilon(t-2) - \varepsilon(t-3)$

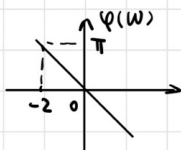
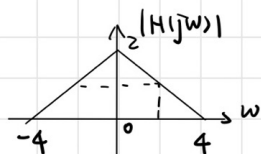
5. 叙述并证傅里叶时域卷积定理

6. 写出  $f(t)$  的傅里叶变换



7.  $e(t) = 6 + 4\sin(2t) + 2\cos(4t) + \sin(6t-1)$

求正弦稳态响应  $r(t)$



8.  $f(t) = e^{-2t} \varepsilon(t) + e^t \varepsilon(-t)$ . 画其 Laplace 变换及收敛域

9.  $u(t) = (10 + 2 \cos 2\pi t + \cos 4\pi t) \cos 1000\pi t$

(1) 求载波信号的频谱图, 以及频带宽度

(2) 求单边调制系数

(3)  $u(t)$  通过  $R=10\Omega$  电阻的平均功率, 载波功率, 边频功率.

(4) AM 调制提高输出效率的方法.

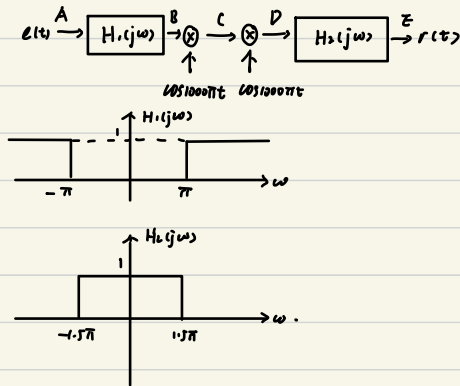
10.  $e(t) = 2 \int a(2\pi t)$  经过如下系统

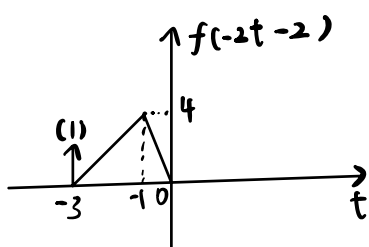
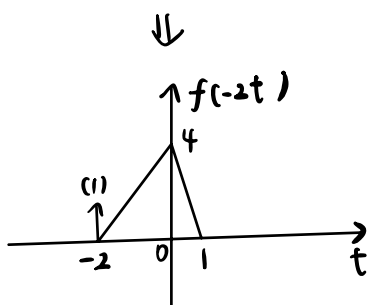
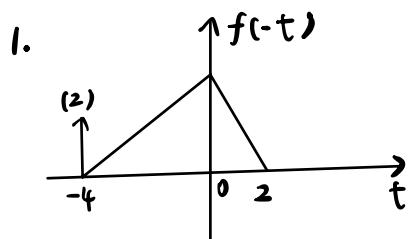
(1) 求  $FT\{e(t)\}$

(2) 作出 A, B, C, D 的频谱图.

(3) 求  $r(t)$

(4) 经系统后, 信号是否失真?





2. 线性非时变

3. ① 求  $r_{zi}(t)$

特征方程:  $(p+3)^2 = 0$

$$\therefore r_{zi}(t) = e^{-3t}(c_1 + c_2 t), \quad t > 0$$

$$\therefore r(0^-) = 1, \quad r'(0^-) = 1$$

$$\therefore \begin{cases} c_1 = 1 \\ -3c_1 + c_2 = 1 \end{cases} \Rightarrow \begin{cases} c_1 = 1 \\ c_2 = 4 \end{cases}$$

$$\therefore r_{zi}(t) = e^{-3t}(1 + 4t) \varepsilon(t)$$

② 求  $h(t)$

$$H(p) = \frac{1}{p+3}$$

$$\therefore h(t) = e^{-3t} \varepsilon(t)$$

4.  $f_1(t) = \varepsilon(t+2) - \varepsilon(t+3)$

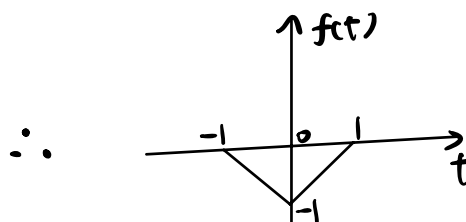
$$f_2(t) = \varepsilon(t-2) - \varepsilon(t-3)$$

易知,  $\varepsilon(t) * \varepsilon(t) = t\varepsilon(t)$

由卷积, 延时性质可知,

$$f_1(t) * f_2(t) = t\varepsilon(t) - (t+1)\varepsilon(t+1) - (t+1)\varepsilon(t+1) + t\varepsilon(t)$$

$$f(t) = (-t-1)[\varepsilon(t+1) - \varepsilon(t)] + (t-1)[\varepsilon(t) - \varepsilon(t-1)]$$



5. 叙述:

$$\text{若 } f_1(t) \leftrightarrow F_1(j\omega)$$

$$f_2(t) \leftrightarrow F_2(j\omega)$$

$$\text{则 } f_1(t) * f_2(t) \leftrightarrow F_1(j\omega) F_2(j\omega)$$

$$\text{证: } F[f_1(t) * f_2(t)]$$

$$= \int_{-\infty}^{+\infty} \left[ \int_{-\infty}^{+\infty} f_1(\tau) \cdot f_2(t-\tau) d\tau \right] e^{-j\omega t} dt$$

$$= \int_{-\infty}^{+\infty} f_1(\tau) \left[ \int_{-\infty}^{+\infty} f_2(t-\tau) e^{-j\omega t} dt \right] d\tau$$

$$= \int_{-\infty}^{+\infty} f_1(\tau) \cdot F_2(j\omega) \cdot e^{-j\omega \tau} d\tau$$

$$= F_1(j\omega) \cdot F_2(j\omega)$$

$$6. f'(t) = -[\varepsilon(t+1) - \varepsilon(t)] + \delta(t) + \varepsilon(t) - \varepsilon(t-1) - \delta(t-1)$$

$$\therefore f'(t) \leftrightarrow -\text{Sa}\left(\frac{\omega}{2}\right) \cdot e^{j\frac{\omega}{2}} + 1 + \text{Sa}\left(\frac{\omega}{2}\right) \cdot e^{-j\frac{\omega}{2}} - e^{-j\omega}$$

$$= \text{Sa}\left(\frac{\omega}{2}\right) \cdot (-2j) \sin \frac{\omega}{2} + 1 - e^{-j\omega}$$

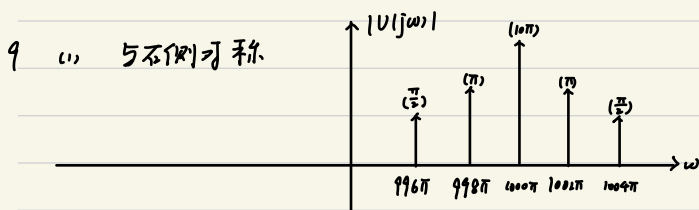
$$\therefore f(t) \leftrightarrow -\text{Sa}^2\left(\frac{\omega}{2}\right) + \frac{1 - e^{-j\omega}}{j\omega}$$

$$\text{或者: } -\text{Sa}^2\left(\frac{\omega}{2}\right) + \text{Sa}\left(\frac{\omega}{2}\right) e^{-j\frac{\omega}{2}}$$

$$7. \text{易得, } r(t) = 12 + 4 \sin(2t - \pi)$$

$$= 12 - 4 \cos\left(2t - \frac{\pi}{2}\right)$$

$$8. \mathcal{L}\{f(t)\} = \frac{1}{s+2} - \frac{1}{s-1} \quad -2 < \operatorname{Re}(s) < 1$$



$$BW = 8\pi \text{ rad/s}$$

$$(2) m_1 = 0.2 \quad m_2 = 0.1$$

$$(3) \bar{P} = \frac{A_0^2}{2R} + \frac{A_0^2}{4R} (m_1^2 + m_2^2) = 5.125W$$

$$\bar{P}_C = \frac{A_0^2}{2R} = 5W \quad \bar{P}_S = \frac{A_0^2}{4R} (m_1^2 + m_2^2) = 0.125W$$

(4) 换 AM-SC 调制/解调 (反调制)

$$10. (1) \mathcal{F}\{e(t)\} = G_{4\pi}(\omega)$$

$$(3) R(j\omega) = \frac{1}{2} (G_{0.5\pi}(\omega + 1.25\pi) + G_{0.5\pi}(\omega - 1.25\pi))$$

$$= \frac{1}{2\pi} G_{0.5\pi}(\omega) * \int \pi [\delta(\omega - 1.25\pi) + \delta(\omega + 1.25\pi)]$$

$$\therefore r(t) = \frac{1}{4} \text{Sa}(0.25\pi t) \cos 1.25\pi t$$

(4) 信号失真

